

CCES 2026

Chula Computer Engineering Showcase

Friday, 15 May 2026 | Sala Phra Kieo



BUSINESS DOMAIN

Enterprise Systems & General Business Platforms

Databank for Cocoa Supply Chain

โกโก้ชาวไทย หัวใจคาเคา

ABSTRACT

Agriculture is a cornerstone of Thailand's economy; however, while staples like rice and sugarcane thrive, cacao cultivation is a nascent industry. Emerging only five years ago, local cacao farming suffers from a significant knowledge gap due to its novelty. There is an urgent need for systems that provide farmers with rapid feedback loops and streamlined data analytics. This project introduces a platform designed to deliver these essential functionalities, bridging the information gap and directly empowering local farmers to optimize their yields.

KEY FEATURES

01 Cocoa Databank Mobile Application

Farmers, Processors and Hub Collectors can enter their Cocoa planting information in the app and it will be sent to a team of researchers to conduct research on better Cocoa planting techniques

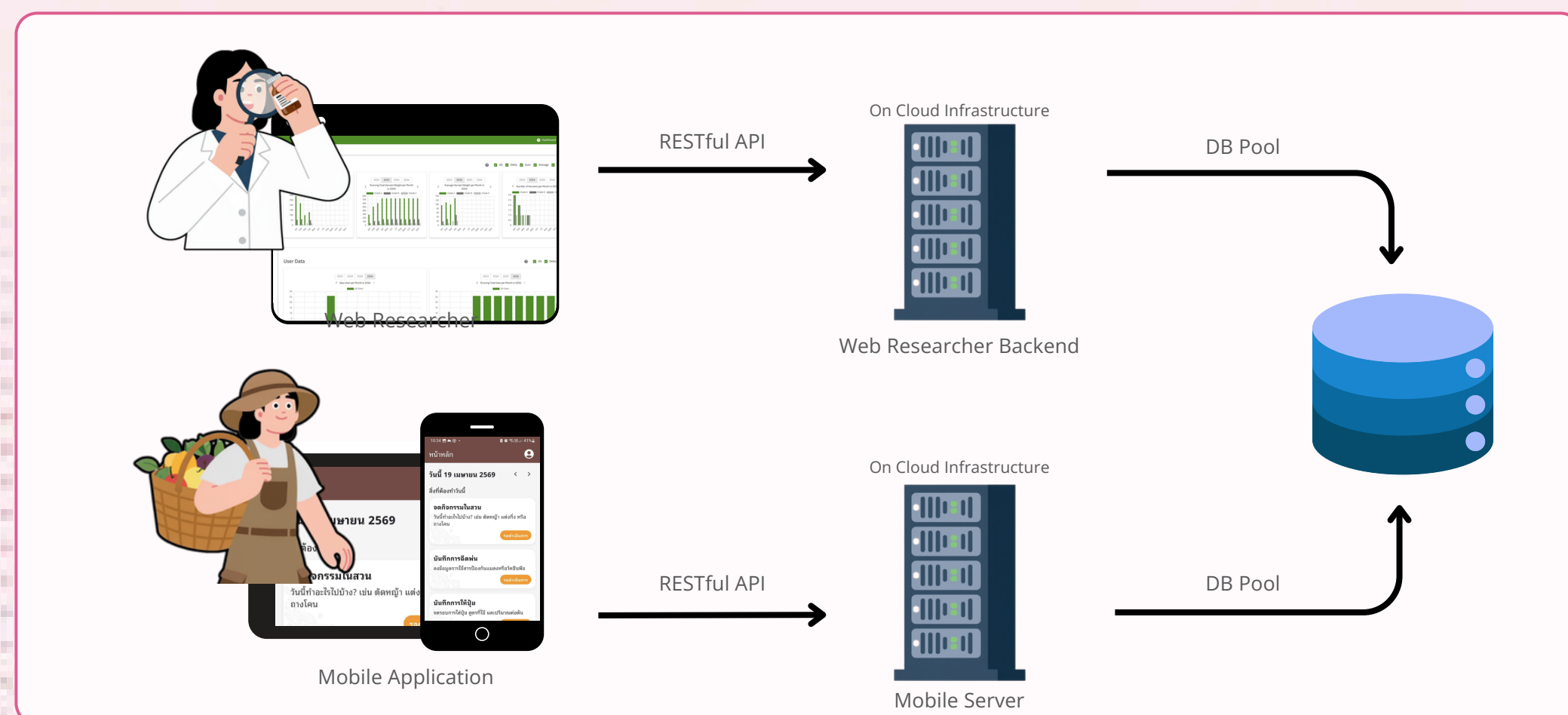
02 Simplified UX/UI for The Elderly

As most in the Cocoa industry are old, the app is optimized to be used by older users, with bigger font size, more comforting colors, easy UI and more

03 Web Researcher (Web Application)

Researchers can view the information sent by the mobile users via the web application. The Form View/Edit function lets the researcher view and edit forms while they can also view the dashboard to view app statistics

SYSTEM OVERVIEW



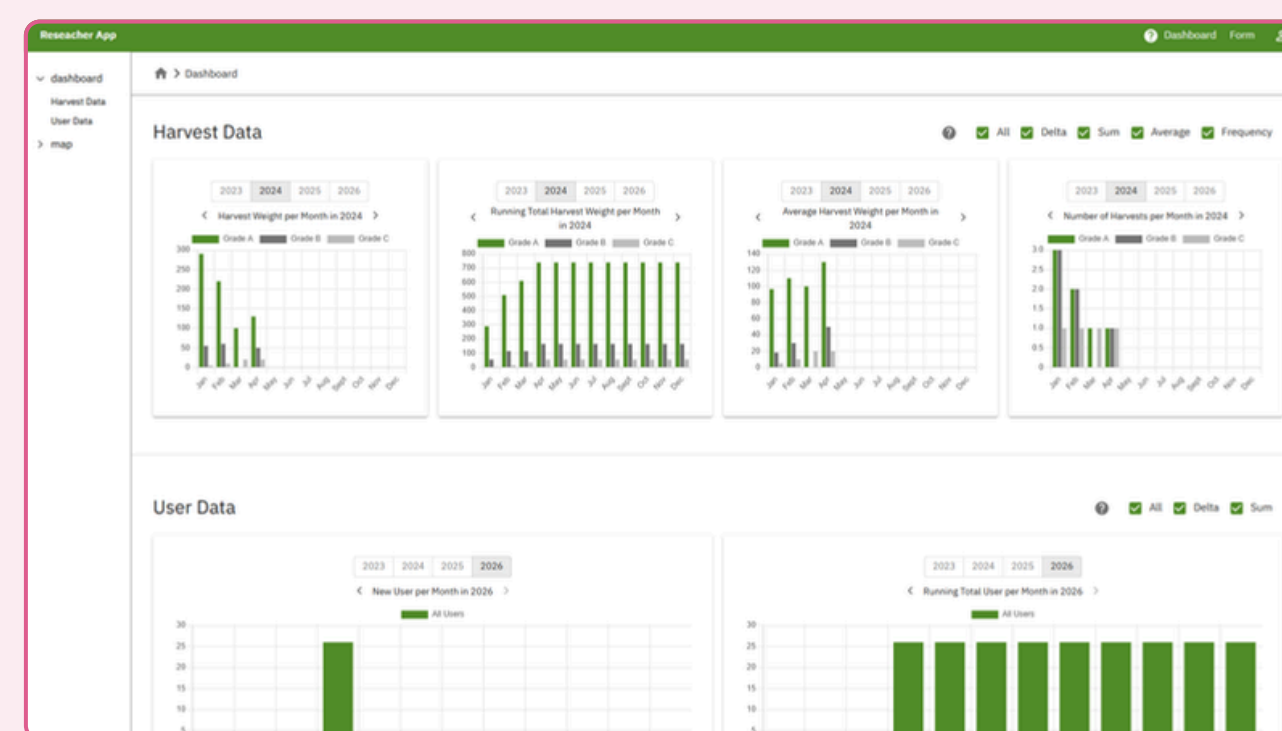
RESULTS

100%
Perceived Utility

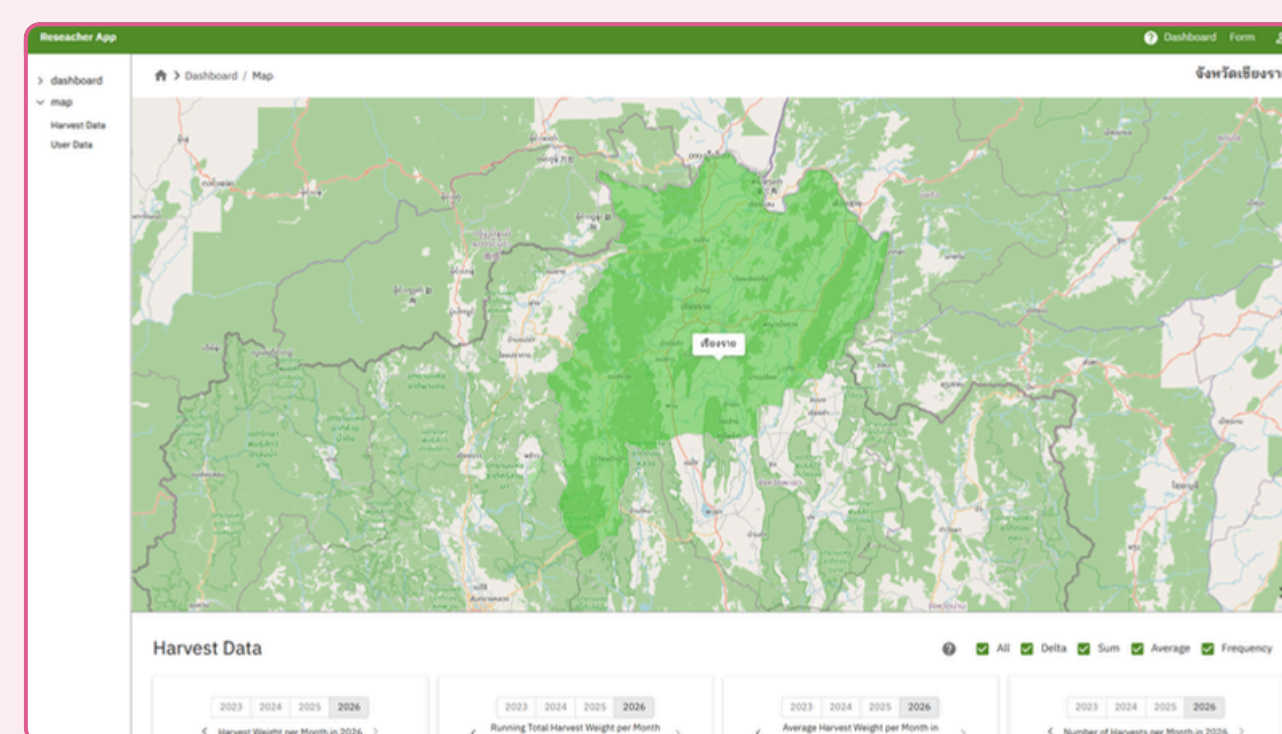
87.5%
High Satisfaction Rate

75%
Operation Smooth

SAMPLE OUTPUTS



Web Researcher - Dashboard



Web Researcher - Provincial Dashboard



Mobile Application - Home Screen

A2 LANDSCAPE (594 × 420 mm)

TEAM MEMBERS

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ADVISOR

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TECHNOLOGIES

Flutter, PostgreSQL, PostGIS,
Spring Boot, Go (Golang), Next.js,
MapLibre, Amazon Web Services
(AWS)

KNOWLEDGE AREA

Software Engineering
System Design
Software Architecture
Database Design
Cloud Computing